Ye Yuan

Staff Research Scientist & Tech Lead, NVIDIA Research

ROBOT LEARNING / HUMANOIDS / EMBODIED AI

yyuan2@cs.cmu.edu • ye-yuan.com • Google Scholar • GitHub

Research profile

Robotics researcher and technical leader with a research record spanning 2017–2026. Develops humanoid learning systems that connect reinforcement learning, visuomotor control, human data, and generative models. Research spans foundational algorithms, large-scale training, and real-robot evaluation. Record includes 47 accepted or published conference/journal papers, nine oral/highlight/spotlight selections, and a SIGGRAPH Best Paper Honorable Mention.

Selected robotics research contributions

Scalable reinforcement learning and whole-body control

Co-first author of SONIC (*Science Robotics*, 2026): scaled motion tracking to 42M parameters, 100M+ motion frames, and 21K GPU-hours, with a shared control interface for teleoperation and vision-language-action models. Adopted across robotics companies and research labs, including Solo Tech, Asratec, and USC's Physical Superintelligence Lab. Builds on first-author Residual Force Control (NeurIPS 2020) and Transform2Act (ICLR 2022 Oral).

Visuomotor learning and sim-to-real transfer

Contributed to VIRAL (CVPR 2026), scaling visual teacher-student training to 64 GPUs and demonstrating 54 continuous zero-shot loco-manipulation cycles on a physical Unitree G1. Related visual RL research studies active perception and dexterity in simulated humanoids.

Learning robot skills from human video and world models

Project lead for GRAIL (CoRL 2026 Oral), using video foundation models in fully specified 3D scenes to generate 20K+ robot demonstrations. Early GR00T fine-tuning experiments with 95% generated data improved grasping success over teleoperation-only training; RGB policies transferred to a physical G1. This program builds on first-author Ego-Pose (2018), SimPoE (CVPR 2021 Oral), and GLAMR (CVPR 2022 Oral).

Generative motion and 3D foundations

Main model architect for Kimodo, a controllable motion diffusion model trained on 700 hours of motion capture. First author of PhysDiff (ICCV 2023 Oral) and last author of GEM (formerly GENMO) (ICCV 2025 Highlight). Co-first author of GAvatar; additional 3D generation work includes AGG and AdaHuman.

Research experience

NVIDIA Research May 2022 – Present

Staff Research Scientist & Tech Lead, DAIR and LPR

Lead research connecting human data, simulation, and generative models to humanoid control and visual policies. Guide collaborations from problem formulation and model design through training, open-source release, and real-robot evaluation.

Carnegie Mellon University, Robotics Institute Aug 2017 – May 2022

Ph.D. Research Assistant; advisor: Kris Kitani

Developed physics-based control, human-motion perception, and generative learning methods. Mentored graduate and undergraduate researchers.

Earlier appointments: NVIDIA AI (2021); Facebook Reality Labs (2018, 2020); Disney Research (2017), research internships.

Selected publications

*Co-first/equal contribution; †project lead. Complete record: full research CV.

Humanoid control and reinforcement learning

1. SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control
   Z. Luo*, **Y. Yuan***, T. Wang*, et al. | Science Robotics 2026
2. Transform2Act: Learning a Transform-and-Control Policy for Efficient Agent Design
   Y. Yuan, Y. Song, Z. Luo, W. Sun, K. Kitani | ICLR 2022; Oral
3. Residual Force Control for Agile Human Behavior Imitation and Extended Motion Synthesis
   Y. Yuan, K. Kitani | NeurIPS 2020

Visuomotor learning and sim-to-real

4. VIRAL: Visual Sim-to-Real at Scale for Humanoid Loco-Manipulation
   T. He*, Z. Wang*, H. Xue*, …, **Y. Yuan**, et al. | CVPR 2026
5. Emergent Active Perception and Dexterity of Simulated Humanoids from Visual Reinforcement Learning
   Z. Luo, C. Tessler, T. Lin, **Y. Yuan**, et al. | arXiv 2025

Human data for robot learning

6. GRAIL: Generating Humanoid Loco-Manipulation from World Models
   T. Xie*, H. Zhang*, J. Park*, …, **Y. Yuan**† | CoRL (accepted) 2026; Oral
7. GLAMR: Global Occlusion-Aware Human Mesh Recovery with Dynamic Cameras
   Y. Yuan, U. Iqbal, P. Molchanov, K. Kitani, J. Kautz | CVPR 2022; Oral
8. SimPoE: Simulated Character Control for 3D Human Pose Estimation
   Y. Yuan, S.-E. Wei, T. Simon, K. Kitani, J. Saragih | CVPR 2021; Oral

Generative motion and 3D modeling

9. Kimodo: Scaling Controllable Human Motion Generation
   D. Rempe*, M. Petrovich*, **Y. Yuan**, et al. | arXiv 2026
10. AdaHuman: Animatable Detailed 3D Human Generation with Compositional Multiview Diffusion
    Y. Huang, **Y. Yuan**, X. Li, J. Kautz, U. Iqbal | ICCV 2025
11. GAvatar: Animatable 3D Gaussian Avatars with Implicit Mesh Learning
    Y. Yuan*, X. Li*, Y. Huang, S. De Mello, K. Nagano, J. Kautz, U. Iqbal | CVPR 2024; Highlight
12. PhysDiff: Physics-Guided Human Motion Diffusion Model
    Y. Yuan, J. Song, U. Iqbal, A. Vahdat, J. Kautz | ICCV 2023; Oral

Education

Carnegie Mellon University, Ph.D. Robotics; advisor: Kris Kitani	2022
Carnegie Mellon University, M.S. Computer Science; advisor: Stelian Coros	2016
Zhejiang University, B.E. Computer Science and Technology	2015

Honors and research leadership

SIGGRAPH Best Paper Honorable Mention (2023); NVIDIA Graduate Fellowship (2021–2022, 1 of 5 selected worldwide); Qualcomm Innovation Fellowship (2020–2021, $100K award). Outstanding Reviewer, ECCV (2022) and ICLR (2021).

Area Chair: CVPR.
Reviewer: NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, ICRA, SIGGRAPH; JMLR, TMLR, TPAMI, TIP, RA-L.

Research software and systems

Project releases: SONIC, GRAIL, GEM/GENMO, GEM-X. **Personal projects:** PyTorch-RL, AgentFormer, RFC, DLow.
Systems: Python, PyTorch, distributed GPU training, Isaac Gym / Isaac Sim, MuJoCo, Unitree G1.